

VLADIMIR, Russian Federation

WMO: 275320

Lat: 56.117N

Lon: 40.350E

Elev: 564

StdP: 14.40

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.3	-7.6	-18.5	2.0	-12.9	-12.5	2.9	-7.2	21.0	23.3	18.9	24.4	5.1	330	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	85.4	69.7	81.8	68.3	78.6	66.8	72.2	81.8	69.9	79.3	67.9	76.4	6.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.8	108.1	78.4	66.4	99.2	75.4	64.5	92.8	73.3	36.2	81.5	34.2	79.4	32.6	76.6	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.7	14.8	DB	-18.6	89.2	6.5	4.5	-23.2	92.4	-27.0	95.1	-30.7	97.6	-35.4	100.8	(4)
(5)			WB	-18.9	75.0	6.4	3.7	-23.5	77.7	-27.2	79.8	-30.8	81.9	-35.5	84.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.4	16.3	17.6	27.5	43.1	55.2	61.9	66.9	62.7	52.7	41.0	28.7	21.1		
	DBStd	20.10	12.45	12.43	8.71	9.28	9.16	7.42	6.67	6.95	7.28	8.16	9.61	11.64		
	HDD50	4835	1046	906	697	247	51	3	1	1	51	296	639	897		
	HDD65	8874	1511	1326	1161	656	324	146	57	126	373	743	1089	1362		
	CDD50	1687	0	0	0	42	213	360	525	396	133	18	0	0		
	CDD65	249	0	0	0	0	20	53	116	56	4	0	0	0		
	CDH74	1998	0	0	0	4	178	433	918	445	20	0	0	0		
	CDH80	528	0	0	0	0	37	106	257	128	0	0	0	0		
	WSAvg	7.1	7.6	7.7	7.7	7.1	7.3	6.6	5.6	5.9	6.3	7.6	7.6	7.9		
	PrecAvg	24.3	1.7	1.3	1.3	1.5	1.9	3.2	2.6	2.4	2.2	2.5	1.9	1.8		
(15) Precipitation	PrecMax	31.5	3.6	2.9	2.8	3.1	3.8	6.9	6.3	5.1	7.4	7.1	4.5	3.6		
	PrecMin	18.5	0.8	0.4	0.2	0.6	0.4	0.3	0.2	0.7	0.5	0.7	0.2	0.7		
	PrecStd	3.5	0.7	0.5	0.6	0.6	0.9	1.6	1.5	1.3	1.4	1.6	1.0	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.7	38.2	53.3	73.7	84.2	87.5	90.9	89.4	76.8	65.5	51.1	43.4		
		MCWB	36.0	34.6	44.7	57.9	66.7	68.3	72.6	73.9	64.8	57.9	48.3	42.1		
	2%	DB	34.8	35.7	45.9	68.1	78.8	82.9	85.7	83.1	72.2	59.4	45.7	39.3		
		MCWB	33.4	33.3	39.2	54.7	64.8	67.7	69.9	70.1	61.9	53.5	44.0	37.9		
	5%	DB	33.4	34.2	41.4	63.0	74.1	79.2	82.4	78.4	67.8	55.3	42.8	36.3		
		MCWB	32.3	32.6	36.4	52.6	61.9	66.1	69.1	67.3	59.6	51.6	41.2	34.9		
	10%	DB	32.0	32.9	38.1	58.6	70.4	75.0	79.1	74.7	64.2	52.1	40.0	33.8		
		MCWB	31.1	31.6	34.6	49.9	59.4	64.2	67.8	65.1	57.7	49.0	38.6	32.7		
	0.4%	WB	36.3	35.9	45.7	59.3	68.8	71.7	76.3	75.7	66.5	58.9	48.9	42.4		
		MCDB	37.1	37.4	52.5	71.0	81.2	83.3	86.5	87.2	73.8	64.5	51.1	43.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.8	34.0	40.2	55.8	65.8	69.5	73.3	71.4	63.3	54.7	44.5	38.2		
		MCDB	34.5	34.8	44.7	66.1	77.0	80.5	82.8	80.5	70.3	57.8	45.6	39.2		
	5%	WB	32.5	32.9	37.1	53.1	63.1	67.3	70.6	68.6	60.8	51.8	41.3	35.0		
		MCDB	33.3	33.9	40.2	61.8	72.6	76.6	79.5	76.8	66.1	54.9	42.4	35.9		
	10%	WB	31.1	31.8	35.2	50.4	60.4	65.0	68.6	66.0	58.2	49.3	38.7	33.0		
		MCDB	32.0	33.0	37.7	58.0	68.5	73.2	77.0	72.8	63.4	51.9	39.9	33.7		
	5% DB	MDBR	9.3	10.7	12.2	15.9	17.8	16.4	16.8	16.1	14.2	9.9	7.0	7.8		
		MCDBR	6.8	8.0	14.9	21.4	22.6	21.5	21.4	21.8	19.5	15.2	9.0	6.6		
		MCWBR	6.8	7.2	10.1	12.1	11.7	10.0	10.1	10.8	11.2	10.5	8.4	6.3		
		MCWBR	7.2	7.1	12.9	20.4	21.6	19.9	19.4	20.4	17.7	13.4	8.3	6.5		
(35) Mean Daily Temperature Range	5% WB	MCWBR	7.2	6.9	9.3	12.1	12.0	10.4	11.0	11.9	11.2	10.1	8.0	6.4		
	taub		0.285	0.325	0.348	0.409	0.387	0.376	0.413	0.427	0.374	0.355	0.316	0.282		
		taud	2.091	2.047	2.072	2.128	2.324	2.379	2.296	2.237	2.364	2.330	2.216	2.087		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		186	223	252	254	270	275	261	247	246	217	175	154		
		Edh at Noon	19	30	38	42	37	36	38	38	29	23	17	15		
	RadAvg		182	463	955	1272	1712	1734	1742	1383	856	395	151	101		
(45) All-Sky Solar Radiation	RadStd		36	75	102	125	127	171	181	147	133	68	30	27		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	+1.31	N/A	N/A	N/A	N/A	N/A	N/A	-360	N/A	N/A

Nomenclature: See separate page